

1. Klausur PraMa

Gruppe B

Beispiel 1

$$1) \quad f(x) = \ln(\cosh(1-x)) \quad x_0 = 1$$

$$f(1) = \ln(\cosh(1-1))$$

$$f(1) = \ln(1)$$

$$f(1) = 0$$

$$f'(x) = \frac{1}{\cosh(1-x)} \cdot \sinh(1-x) \cdot (-1)$$

$$f'(x) = - \frac{1}{\cosh(1-x)} \cdot \sinh(1-x)$$

$$f'(1) = - \frac{1}{\cosh(1-1)} \cdot \sinh(1-1)$$

$$f'(1) = - \frac{1}{\cosh(0)} \cdot \sinh(0)$$

$$f'(1) = - \frac{1}{1} \cdot 0$$

$$f'(1) = 0$$

$$f'(x) = -\frac{\sinh(1-x)}{\cosh(1-x)} \leftarrow f_1$$

$$\cosh(1-x) \leftarrow f_2$$

$$f''(x) = \frac{f_1' \cdot f_2 - f_1 \cdot f_2'}{f_2^2}$$

$$f_1 = \sinh(1-x)$$

$$f_1' = (\sinh(1-x))' = \cosh(1-x) \cdot (-1) = -\cosh(1-x)$$

$$f_2 = \cosh(1-x)$$

$$f_2' = (\cosh(1-x))' = \sinh(1-x) \cdot (-1) = -\sinh(1-x)$$

$$f''(x) = -\frac{(-\cosh(1-x) \cdot \cosh(1-x)) - (\sinh(1-x) \cdot (-\sinh(1-x)))}{\cosh^2(1-x)}$$

$$f''(1) = -\frac{(-\cosh(1-1) \cdot \cosh(1-1)) - (\sinh(1-1) \cdot (-\sinh(1-1)))}{\cosh^2(1-1)}$$

$$f''(1) = -\frac{(-\cosh(0) \cdot \cosh(0)) - (\sinh(0) \cdot (-\sinh(0)))}{\cosh^2(0)}$$

$$f''(1) = -\frac{(-1 \cdot 1) - (0 \cdot (-0))}{1}$$

$$f''(1) = -\frac{(-1)}{1}$$

$$f''(1) = 1$$

$$T_2(1) = \frac{0}{0!} \cdot (x-1)^0 + \frac{0}{1!} \cdot (x-1)^1 + \frac{1}{2!} \cdot (x-1)^2$$

$$\underline{\text{NR:}} \quad \frac{1}{2} \cdot (x-1) \cdot (x-1) = \frac{1}{2} \cdot (x^2 - 2x + 1) = \frac{x^2}{2} - x + \frac{1}{2}$$

$$T_2(1) = \frac{1}{2}x^2 - x + \frac{1}{2}$$

Beispiel 2

Daniel Köstler
12307040

$$f(x) = x \cdot \sqrt{1-x^2}$$

$$\begin{aligned}\cos^2 t + \sin^2 t &= 1 \\ \cos^2 t &= 1 - \sin^2 t\end{aligned}$$

Spezielle Substitution:

$$F(x) = \int x \cdot \sqrt{1-x^2} dx = \left| \begin{array}{l} x = \sin(u) \\ dx = \cos(u) du \end{array} \right| =$$

$$\int \sin(u) \cdot \sqrt{1-\sin^2 u} \cos(u) du = \int \sin(u) \cdot \cos^2(u) du =$$

$$\int \sin(u) du = -\cos u$$

$$(\cos^2 u)' = -2 \sin u$$

Normale Subst.

$$f(x) = x \cdot \sqrt{1-x^2}$$

$$f(x) = x \cdot (1-x^2)^{\frac{1}{2}}$$

$$\int x \cdot (1-x^2)^{\frac{1}{2}} dx = \left| \begin{array}{l} u = 1-x^2 \\ du = -2x dx \\ -\frac{1}{2} du = dx \end{array} \right| = -\frac{1}{2} \cdot \int \frac{x}{x} \cdot u^{\frac{1}{2}} du = -\frac{1}{2} \cdot \int u^{\frac{1}{2}} du$$

$$= -\frac{1}{2} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}} = -\frac{1}{2} \cdot \frac{2}{3} \cdot u^{\frac{3}{2}}$$

Rücksubst.

$$= -\frac{2}{6} \cdot (1-x^2)^{\frac{3}{2}}$$

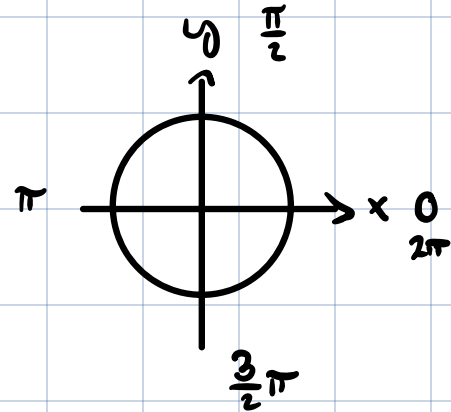
Beispiel 3

Daniel Köstler

12307040

$$C = \{ \vec{r}(t) = \begin{pmatrix} 2 + \frac{1}{5}t \\ 3 + \frac{1}{5}t \end{pmatrix} : t \in [-1, 0] \}$$

$$f(x, y) = x^2 \cos(x) + e^y + y \cdot \sin(x)$$



a) ges. Tangentialebene $(x_0, y_0) = \begin{pmatrix} 2\pi \\ 5 \end{pmatrix}$

$$\vec{T}(x_0, y_0) = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot ((x, y) - (x_0, y_0))$$

$$f(2\pi, 5) = (2\pi)^2 \cdot (\cos(2\pi)) + e^5 + 5 \cdot \sin(2\pi)$$

$$f(2\pi, 5) = 4\pi^2 \cdot 1 + e^5 + 5 \cdot 0$$

$$f(2\pi, 5) = 4\pi^2 + e^5$$

$$f(x, y) = x^2 \cos(x) + e^y + y \cdot \sin(x)$$

$$\frac{\partial f}{\partial x} = 2x \cdot \cos(x) - x^2 \cdot \sin(x) + \cos(x) \quad \frac{\partial f}{\partial y} = e^y + \sin(x)$$

NR: $f_1' \cdot f_2 + f_1 \cdot f_2'$

$$(x^2 \cos(x))' = 2x \cdot \cos(x) - x^2 \cdot \sin(x)$$

$$(x^2)' = 2x$$

$$(\cos x)' = -\sin(x)$$

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

$$\nabla f(x, y) = \begin{pmatrix} 2x \cdot \cos(x) - x^2 \cdot \sin(x) + \cos(x) \\ e^y + \sin(x) \end{pmatrix}$$

$$\nabla f(2\pi, 5) = \begin{pmatrix} 4\pi \cdot \cos(2\pi) - 4\pi^2 \cdot \sin(2\pi) + \cos(2\pi) \\ e^5 + \sin(2\pi) \end{pmatrix} = \begin{pmatrix} 4\pi \cdot 1 - 0 + 1 \\ e^5 + 0 \end{pmatrix} = \begin{pmatrix} 4\pi + 1 \\ e^5 \end{pmatrix}$$

$$\vec{T}(2\pi, 5) = (4\pi^2 + e^5) + \begin{pmatrix} 4\pi+1 \\ e^5 \end{pmatrix} \cdot \begin{pmatrix} x-2\pi \\ y-5 \end{pmatrix}$$

NR: $(4\pi+1) \cdot (x-2\pi) = 4\pi x - 8\pi^2 + x - 2\pi$

$$e^5 \cdot (y-5) = e^5 y - 5e^5$$

$$\vec{T}(2\pi, 5) = 4\pi^2 + e^5 + 4\pi x - 8\pi^2 + x - 2\pi + e^5 y - 5e^5$$

$$\vec{T}(2\pi, 5) = -4\pi^2 - 4e^5 + 4\pi x + x - 2\pi + e^5 y$$

b) ges. $s(t)$ und L

$$s(t) = \int_0^t ds = \int_0^t |d\vec{r}| = \int_0^t |\dot{\vec{r}}(t)| dt$$

$$\vec{r}(t) = \begin{pmatrix} 2t-5 \\ 3+\frac{1}{5}t \end{pmatrix} \quad \dot{\vec{r}}(t) = \begin{pmatrix} 2 \\ \frac{1}{5} \end{pmatrix} \quad |\dot{\vec{r}}(t)| = \sqrt{2^2 + \left(\frac{1}{5}\right)^2} = \sqrt{4 + \frac{1}{25}}$$

$$= \sqrt{\frac{100}{25} + \frac{1}{25}} = \sqrt{\frac{101}{25}}$$

$$s(t) = \int_{-1}^t \sqrt{\frac{101}{25}} d\tau = \sqrt{\frac{101}{25}} \tau \Big|_{\tau=-1}^{\tau=t} = \sqrt{\frac{101}{25}} t - \left(-\sqrt{\frac{101}{25}}\right) = \sqrt{\frac{101}{25}} t + \sqrt{\frac{101}{25}}$$

$$t=0$$

$$L = \sqrt{\frac{101}{25}} \cdot 0 + \sqrt{\frac{101}{25}} = \sqrt{\frac{101}{25}}$$

Beispiel 4

Daniel Köstler
12307040

$$\vec{f}(x, y, z) = \begin{pmatrix} \ln(xy) + 1 \\ \frac{x}{y} - z \cdot \cos(yz) \\ f_3(x, y, z) \end{pmatrix}$$

$$\frac{\partial \Phi}{\partial x} = \ln(xy) + 1$$

$$\int \ln(xy) + 1 dx = \left| \begin{array}{l} u = xy \\ du = y dx \\ \frac{1}{y} du = dx \end{array} \right| = \frac{1}{y} \cdot \int \ln(u) + 1 du$$

$$= \frac{1}{y} \cdot (u \cdot \ln(u) - u + u) = \frac{1}{y} \cdot (u \ln(u)) + c(y, z)$$

Rücksubstituieren:

$$\frac{1}{y} \cdot ((xy) \cdot \ln(xy)) = \frac{xy \cdot \ln(xy)}{y} + c(y, z)$$

$$\frac{\partial \Phi}{\partial y} \frac{xy \cdot \ln(xy)}{y} + c(y, z) = \frac{xy}{y^2} + \frac{\partial c}{\partial y} = \frac{x}{y} + \frac{\partial c}{\partial y}$$

$$\text{NR: } \frac{f_1' \cdot f_2 - f_1 \cdot f_2'}{f_2'}$$

$$\frac{xy \cdot \ln(xy)}{y} = \frac{(x \cdot \ln(xy) + x) \cdot y - xy \cdot \ln(xy)}{y^2} = \frac{xy \cdot \ln(xy) + xy - xy \cdot \ln(xy)}{y^2}$$

$$f_1 = xy \cdot \ln(xy)$$

$$f_1' = x \cdot \ln(xy) + xy \cdot \frac{1}{xy} \cdot x = x \cdot \ln(xy) + x$$

$$f_2 = y$$

$$f_2' = 1$$

$$\frac{\partial c}{\partial y} = -z \cos(yz)$$

$$\int -z \cdot \cos(yz) dy = z \cdot \sin(yz) \cdot z = z^2 \cdot \sin(yz) + c(f_3)$$

$$\Phi = \frac{x}{y} - z \cdot \cos(yz)$$

$$\frac{\partial \Phi}{\partial z} \frac{x}{y} - z \cdot \cos(yz) = -\cos(yz) + \sin(yz) \cdot z$$

NR:

$$-z \cdot \cos(yz) = -1 \cdot \cos(yz) + (-1) \cdot (-\sin yz) \cdot z$$

$$(-z)' = -1 \quad = -\cos(yz) + \sin yz \cdot z$$

$$\cos(yz) = -\sin(yz) \cdot z$$

Einheitsvektor $\vec{a}_0$:

$$\frac{1}{|\vec{a}|} \cdot \vec{a}$$

Kartesische (kanonische) Basis:

$$\vec{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\forall \vec{e}_x \cdot \forall \vec{e}_y = \vec{0}$$

Länge Vektor $\vec{a} = \begin{pmatrix} x \\ y \end{pmatrix}$:

$$|\vec{a}| = \sqrt{x^2 + y^2}$$

Zylinderkoordinaten:

$$x = r \cdot \cos \varphi \quad \varphi = \arctan\left(\frac{y}{x}\right)$$

$$y = r \cdot \sin \varphi \quad r = \sqrt{x^2 + y^2}$$

Kugelkoordinaten:

$$x = r \cdot \sin \Theta \cdot \cos \varphi \quad \Theta = \arccos\left(\frac{z}{r}\right)$$

$$y = r \cdot \sin \Theta \cdot \sin \varphi \quad \varphi = \arctan\left(\frac{y}{x}\right)$$

$$z = r \cdot \cos \Theta$$

Tangentengleichung:

$$t(x) = f(x_0) + f'(x_0) \cdot (x - x_0)$$

Integrationen:

$$\int \tan x \, dx = -\ln |\cos x|$$

$$\int \ln x \, dx = x \cdot \ln x - x$$

Spezielle Substitutionen:

$$\int R(\sin x, \cos x) \, dx \Rightarrow t = \tan\left(\frac{x}{2}\right)$$

$$\int R(\sqrt{1+x^2}) \, dx \Rightarrow x = \sinh(t)$$

$$\int R(\sqrt{x^2-1}) \, dx \Rightarrow x = \cosh(t)$$

$$\int R(\sqrt{1-x^2}) \, dx \Rightarrow x = \sin(t) / \cos(t)$$

Ableitungen:

$$(\cot x)' = -\frac{1}{\sin^2 x}$$

$$(\ln x)' = \frac{1}{x}$$

$$(\tanh x)' = \frac{1}{\cosh^2 x}$$

$$(\sin(x))' = \cos(x)$$

$$(\cos(x))' = -\sin(x)$$

Ableitung der Umkehrfunktion:

$$(f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))}$$

Taylorpolynom:

$$\frac{f(a)}{0!} \cdot (x-a)^0 + \frac{f'(a)}{1!} \cdot (x-a)^1 + \frac{f''(a)}{2!} \cdot (x-a)^2 \dots$$

Flächeninhalt zwischen 2 Fkt.:

$$A = \int_{x_1}^{x_2} |f(x) - g(x)| \, dx$$

Rotationsvolumen:

$$\text{um } x\text{-Achse: } \pi \cdot \int_a^b f(x)^2 \, dx$$

$$\text{um } y\text{-Achse: } \pi \cdot \int_a^b (f^{-1}(x))^2 \, dx$$

Komplexe Zahlen:

$$z = |z| \cdot e^{i\varphi} \quad \arctan\left(\frac{b}{a}\right) = \arg(z)$$

$$e^{i\varphi} = \cos(\varphi) + i \cdot \sin(\varphi) \quad |z| = \sqrt{a^2 + b^2}$$

Ableitungsregeln:

$$(f_1 \cdot f_2)' = f_1' \cdot f_2 + f_1 \cdot f_2'$$

$$\left(\frac{f_1}{f_2}\right)' = \frac{f_1' \cdot f_2 - f_1 \cdot f_2'}{f_2^2}$$

Winkel zu Vektoren:

$$\cos \varphi = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \cdot |\vec{b}|}$$

Tangenten- / Tangentialvektor:

$$\vec{T}(t) = \frac{1}{|\dot{\vec{r}}(t)|} \cdot \dot{\vec{r}}(t)$$

Tangente:

$$\vec{r}(s) = \vec{r}(t) + s \cdot \vec{T}(t)$$

Richtungsableitung Skalarfeld:

$$D_{\vec{e}}(f) = \nabla f(\vec{r}) \cdot \hat{e}$$

$\hat{e}$ (Einheitsvektor)

Tangentialebene im Punkt r_0 :

$$\vec{T}(\vec{r}) = \vec{T}(x, y) = f(\vec{r}_0) + \nabla f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0)$$

Vektorfeld:

$\text{rot } \vec{f}(x, y, z) = 0 \dots$ wirbelfrei

$\text{div } \vec{f}(x, y, z) = 0 \dots$ quellenfrei

Bogenlänge:

$$s(t) = \int_a^t ds = \int_a^t |\dot{\vec{r}}(\tau)| d\tau$$

$$ds = |d\vec{r}| = |\dot{\vec{r}}(\tau)| d\tau$$

$$L = \int_a^b ds = \int_a^b |\dot{\vec{r}}(\tau)| d\tau$$

Kurvenintegrale:

Skalarfeld:

$$\int_a^b f(\vec{r}(t)) \cdot |\dot{\vec{r}}(t)| dt$$

Vektorfeld:

$$\int_a^b \vec{f}(\vec{r}(t)) \cdot \dot{\vec{r}}(t) dt$$

$$\ln(1) = 0$$
$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$

Hauptsatz über Kurvenintegrale:

wenn $\text{rot } \vec{f} = 0$, dann:

$$\int_c \vec{f} \cdot d\vec{r} = \Phi(\vec{r}(b)) - \Phi(\vec{r}(a))$$

Geschlossene Kurve:

$$\oint \vec{f} \cdot d\vec{r} = 0$$

Trigonometrie:

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\sin(2x) = 2 \cdot \sin x \cdot \cos x$$

$$\cos(2x) = 1 - 2 \sin^2 x$$

$$\cos(2x) = 2 \cos^2 x - 1$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

Hyperbolische Trig.:

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh x = \frac{1}{2} \cdot (e^x - e^{-x})$$

$$\cosh x = \frac{1}{2} \cdot (e^x + e^{-x})$$

Weitere Regeln:

$$\sin x \cdot \sin y = \frac{1}{2} \cdot (\cos(x-y) - \cos(x+y))$$

$$\cos x \cdot \cos y = \frac{1}{2} \cdot (\cos(x-y) + \cos(x+y))$$

$$\sin x \cdot \cos x = \frac{1}{2} \cdot (\sin(x-y) + \sin(x+y))$$

α	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	-
cot	-	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0